

07. Transport of Calcium, Magnesium, and Phosphate · assets 1-9 / 15

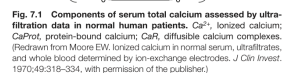


Fig 7 1

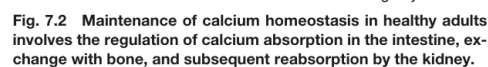


Fig. 7.2 Maintenance of calcium homeostasis in healthy adults involves the regulation of calcium absorption in the intestine, exchange with bone, and subsequent reabsorption by the kidney.

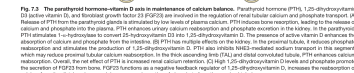


Fig 7 3

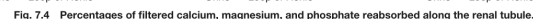


Fig 7 4

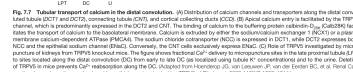
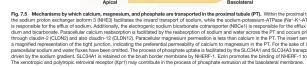
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Fig. 7 Mechanisms by which calcium, magnesium, and phosphate are transported in the proximal tubule (PT). Within the proximal PT, the sodium proton exchanger isoform 3 (NHE3) facilitates the inward transport of sodium, while the sodium-calcium cotransporter (SNC) is responsible for the efflux of calcium. Additionally, the electrogenic sodium-magnesium cotransporter (SMC) is responsible for the efflux of magnesium. The apical membrane of the PT contains two distinct channels for the entry of calcium: the transient receptor potential channel-6 (TRPC6) and also dihydropyridine receptor-related protein-2 (DHPR-2). Paracellular magnesium permeation is less than calcium in the PT. The inset shows a magnified representation of the tight junction, indicating the preferred permeability of calcium to magnesium in the PT. For the most part, the majority of sodium and extrinsic calcium exit the PT via the basolateral membrane. The presence of a chloride channel and SGLCA7-mediated sodium by the basolateral gradient. SGLCA5A1 is selective on the brush border membrane by NHE3F1. Extrn promotes the loading of NaHCO₃. The vacuolar and polytopic retention receptor (Ryr) may contribute in the process of phosphate extrusion at the basolateral membrane.

Fig 7 5

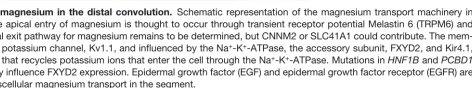


Fig. 7.8 Tubular transport of magnesium in the distal convoluted. Schematic representation of the magnesium transport machinery in the distal convoluted tubule. The apical entry of magnesium is thought to occur through transient receptor potential Melastain 6 (TRPM6) and TRPM7 channels. The basolateral exit pathway for magnesium remains to be determined, but CNNM2 or SLCA1A1 could contribute. The membrane voltage is adjusted by the potassium channel, Kv1.1, and influenced by the Na⁺/K⁺-ATPase, as well as additional subunits, FXND2, and Klrk1, a basolateral potassium channel that recycles potassium ions that enter the cell through the Na⁺/K⁺-ATPase. Mutations in *HNF1B* and *PCBD1* have been associated with hypomagnesemia in the context of the factor (EGF) and epidermal growth factor receptor (EGFR) are involved in the regulation of transcellular magnesium transport in the segment.

Fig_7_8

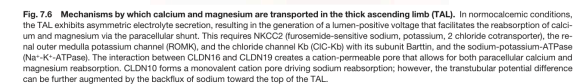


Fig. 7.6 Mechanisms by which calcium and magnesium are transported in the thick ascending limb (TAL). In normocalcemic conditions, the TAL exhibits asymmetric electrolyte secretion, resulting in the generation of a lumen-positive voltage that facilitates the reabsorption of calcium and magnesium via the paracellular shunt. This requires NKCC2 (fluoride-sensitive sodium, potassium, 2 chloride cotransporter), the renal outer apical potassium channel (ROMK), and the chloride channel ClC (CIC-7) with its subunit BTRN1, and the sodium-potassium-ATPase ($\text{Na}^+/\text{K}^+/\text{ATPase}$). The interaction between CLDN16 and CLDN19 creates a cation-permeable pore that allows for both paracellular calcium and magnesium reabsorption. CLDN19 forms a monovalent cation pore driving sodium reabsorption; however, the transtubular potential difference can be further augmented by the backflux of sodium toward the top of the TAL.

Fig 7 6

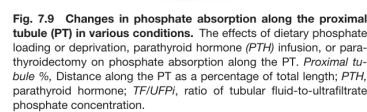


Fig. 7.9 Changes in phosphate absorption along the proximal tubule (PT) in various conditions. The effects of dietary phosphate loading or deprivation, parathyroid hormone (PTH) infusion, or parathyroidectomy on phosphate absorption along the PT. *Proximal tubule %*, Distance along the PT as a percentage of total length; *PTH*, parathyroid hormone; *TF/UUPI*, ratio of tubular fluid-to-ultrafiltrate phosphate concentration.

Fig_7_9

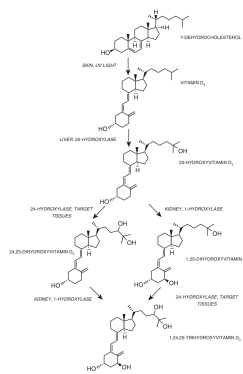


Fig. 7.10 Formation and metabolism of vitamin D3. 7-dehydrocholesterol, 7-dehydrocholesterol, 25-hydroxyvitamin D3, 25-hydroxyvitamin D3, 24-hydroxyvitamin D3, 24-hydroxyvitamin D3, 1,25-dihydroxyvitamin D3, 1,25-dihydroxyvitamin D3, 1,24-dihydroxyvitamin D3, 1,24-dihydroxyvitamin D3.

Fig_7_10

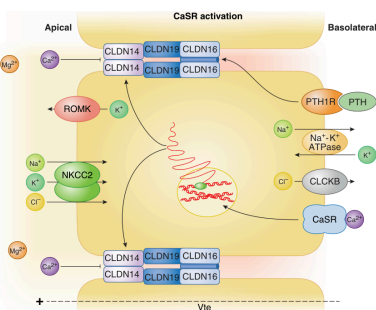


Fig. 7.11 Regulation of calcium transport in the thick ascending limb. Schematic representation of the proposed model for regulation of claudin 14 (CLDN14) by the calcium-sensing receptor (CaSR). Activation of CaSR and subsequent stimulation of CLDN14 expression occurs in the presence of hypercalcemia. CLDN14 insertion into the tight junction complex inhibits the movement of divalent cations across the segment. NKCC2 (furosemide sensitive sodium, potassium, 2 chloride cotransporter), the renal outer medulla potassium channel (ROMK), the chloride channel Kb (ClC-Kb) with its subunit Bartsin, and the sodium-potassium-ATPase (Na+-K+-ATPase) are required for generating the lumen-positive voltage required for divalent cations reabsorption via CLDN16 and CLDN19.

Fig_7_11

Table 7.1 Distribution and Concentrations of Magnesium in a Healthy Adult

Site	% Total-Body Mg	Concentration/Content
Bone	53	0.5% of bone ash
Muscle	27	9mmol/kg wet weight
Soft tissue	19	9mmol/kg wet weight
Adipose tissue	0.012	0.8mmol/kg wet weight
Erythrocytes	0.5	1.65-2.73mmol/L
Serum	0.3	0.69-0.94mmol/L

From Yu ASL, Chertow GM, Luyckx V, Marsden PA, Skorecki K, Taal MW, eds. *Brenner and Rector's The Kidney*. 11th ed. Philadelphia, PA: Elsevier; 2019.

Table_7_1

Table 7.2 Factors Altering the Reabsorption of Calcium in the Kidney

Calcium Reabsorption Increased By	Calcium Reabsorption Decreased By
1,25-dihydroxyvitamin D	Aminoglycosides
Amiloride	Androgens
Calcitonin	Calcimimetics
Estrogen	Calcineurin inhibitors
Fibroblast growth factor 23 Klotho	Hypercalcemia
Hypocalcemia	Insulin/glucose
Metabolic alkalosis	Loop diuretics
Phosphate loading	Metabolic acidosis
PTH	Phosphate depletion
PTH-related peptide	Volume expansion
Thiazides	
Volume contraction	

Table_7_2

Table 7.3 Factors Altering the Reabsorption of Magnesium in the Kidney

Magnesium Reabsorption Increased By	Magnesium Reabsorption Decreased By
β-Adrenergic agonists	1,25-dihydroxyvitamin D
Aldosterone	Aminoglycosides
Amiloride	Calcineurin inhibitors
Arginine vasopressin	Hypercalcemia
Calcitonin	Loop diuretics
Glucagon	Magnesium loading
Insulin	Metabolic acidosis
Magnesium restriction	Phosphate depletion
Metabolic alkalosis	Prostaglandin E ₂
Parathyroid hormone	Thiazide

Table_7_3

Table 7.4 Factors Altering the Reabsorption of Phosphate in the Kidney

Phosphate Reabsorption Increased By	Phosphate Reabsorption Decreased By
Growth hormone	Atrial natriuretic peptide
Hypocalcemia	Diuretics
Insulin	Dopamine
Metabolic alkalosis	Estrogen
Parathyroidectomy	Fibroblast growth factor 7
Phosphate depletion	Fibroblast growth factor 23
Stanniocalcin 1	Glucocorticoids
Volume contraction	Hypercalcemia
	Hypokalemia
	Matrix extracellular phosphoglycoprotein (MEPE)
	Metabolic acidosis
	Parathyroid hormone
	Phosphate loading
	Secreted frizzled-related protein 4 (sFRP-4)
	Volume expansion

Table_7_4